

FINANCE & BANKING ANALYSIS

Power BI Project Documentation

Snowflake • Power BI • DAX • Data Analytics

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1. Project Overview

The Finance & Banking Analysis project is an interactive Business Intelligence solution developed using Microsoft Power BI with Snowflake as the primary database source. The report converts financial transaction, branch, customer, account and loan data into an interactive dashboard that supports financial monitoring and analysis.

1.1 Project Purpose

- Monitor revenue, expenses and profit across the banking business.
- Analyze transaction volume, transaction value and transaction categories.
- Compare financial performance across branches.
- Analyze the loan portfolio by loan type, status, amount and interest rate.
- Provide customer and gender-based deposit analysis.
- Provide interactive drill-through, navigation, filtering and security features.

1.2 Key Outcomes

- Centralized financial reporting in one Power BI report.
- Interactive analysis using slicers and cross-filtering.
- Branch-level drill-through for detailed transaction analysis.
- DAX-based KPIs and time-intelligence calculations.
- Role-based access using Row-Level Security.

2. Business Objectives

Business Area	Objective
Financial Performance	Track revenue, expenses, profit and profit margin.
Transactions	Analyze transaction count, amount, type and direction.
Branches	Compare branch revenue, expense, profit and transaction performance.
Loans	Analyze loan portfolio, loan types, status, amount and interest rate.
Customers	Analyze customer-related financial activity, including gender-wise deposits.
Security	Restrict report data according to the user's assigned branch or access level.

3. Technologies Used

Technology	Purpose
Snowflake	Cloud data warehouse and source database.
Microsoft Power BI	Data modeling, visualization, dashboard development and publishing.
DAX	Measures, KPIs, calculations and time intelligence.
Power Query	Data connection, transformation and preparation.
Power BI Service	Report publishing, sharing and RLS security management.

4. Data Source – Snowflake

The project uses Snowflake as the database source. Data is loaded into Power BI through a Snowflake connection, prepared through Power Query where required, and modeled using a fact-and-dimension structure.

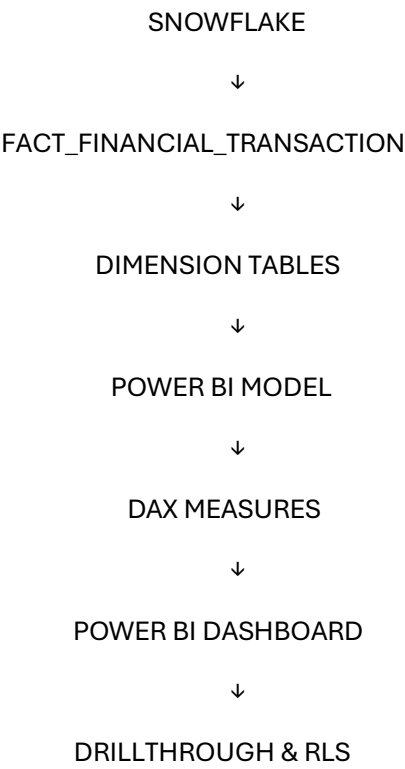
4.1 Main Source Tables

Table	Type	Purpose
FACT_FINANCIAL_TRANSACTION	Fact	Stores financial transaction-level measures.
DIM_DATE	Dimension	Provides date, month, quarter, year and calendar attributes.
DIM_BRANCH	Dimension	Contains branch information used for branch analysis.
DIM_TRANSACTION	Dimension	Contains transaction type, category, direction and flags.
DIM_LOAN	Dimension	Contains loan amount, type, status, rate and term details.
DIM_ACCOUNT	Dimension	Contains account-level information.
DIM_CUSTOMER	Dimension	Contains customer attributes used for customer analysis.

4.2 Data Flow

Snowflake Database → Power Query → Power BI Data Model → DAX Measures → Interactive Report → Power BI Service

5. Architecture



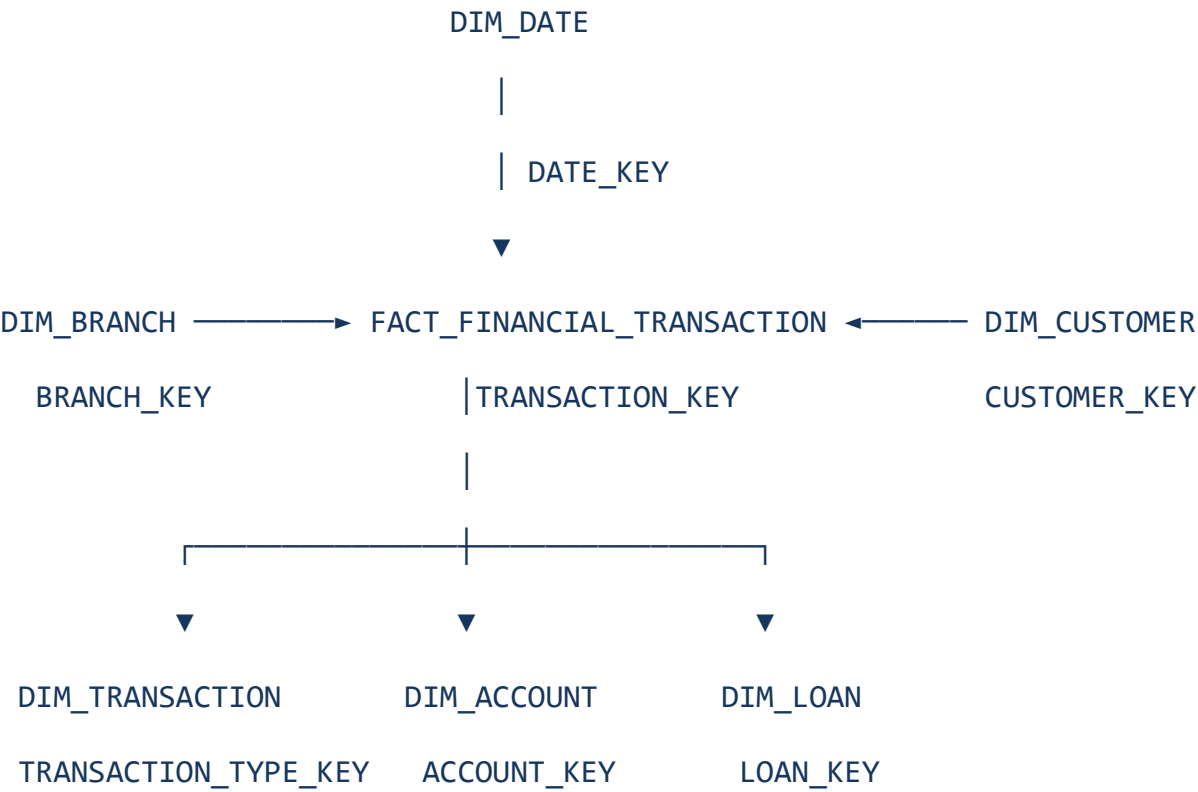
The report follows a star-schema-oriented design. The central financial transaction fact table is connected to supporting dimension tables. This structure improves filtering, aggregation and dashboard performance.

5.1 Model Relationships

Dimension	Fact	Key / Relationship
DIM_DATE	FACT_FINANCIAL_TRANSACTION	DATE_KEY
DIM_BRANCH	FACT_FINANCIAL_TRANSACTION	BRANCH_KEY
DIM_CUSTOMER	FACT_FINANCIAL_TRANSACTION	CUSTOMER_KEY
DIM_ACCOUNT	FACT_FINANCIAL_TRANSACTION	ACCOUNT_KEY
DIM_TRANSACTION	FACT_FINANCIAL_TRANSACTION	TRANSACTION_TYPE_KEY
DIM_LOAN	FACT_FINANCIAL_TRANSACTION	LOAN_KEY

6. ER Diagram / Star Schema

Logical model used for the Power BI report:



7. Power BI Report Pages

Page	Purpose	Main Content
Home	Landing and navigation page.	Navigation cards and key project highlights.
Financial Overview	Overall financial performance.	Revenue, expense, profit, transactions and trends.
Loan Analysis	Loan portfolio analysis.	Loan type, amount, status, rate and trend analysis.

Branch Performance	Branch comparison.	Revenue, expense, profit and transaction performance.
Transaction Analysis	Transaction-level analysis.	Transaction volume, amount, direction, category and type.
Risk Analysis	Loan status and portfolio monitoring.	Loan status, portfolio amount and interest-rate metrics.
Branch Details	Drillthrough detail.	Matrix-based transaction analysis for the selected branch.

7.1 Home Page Screenshot

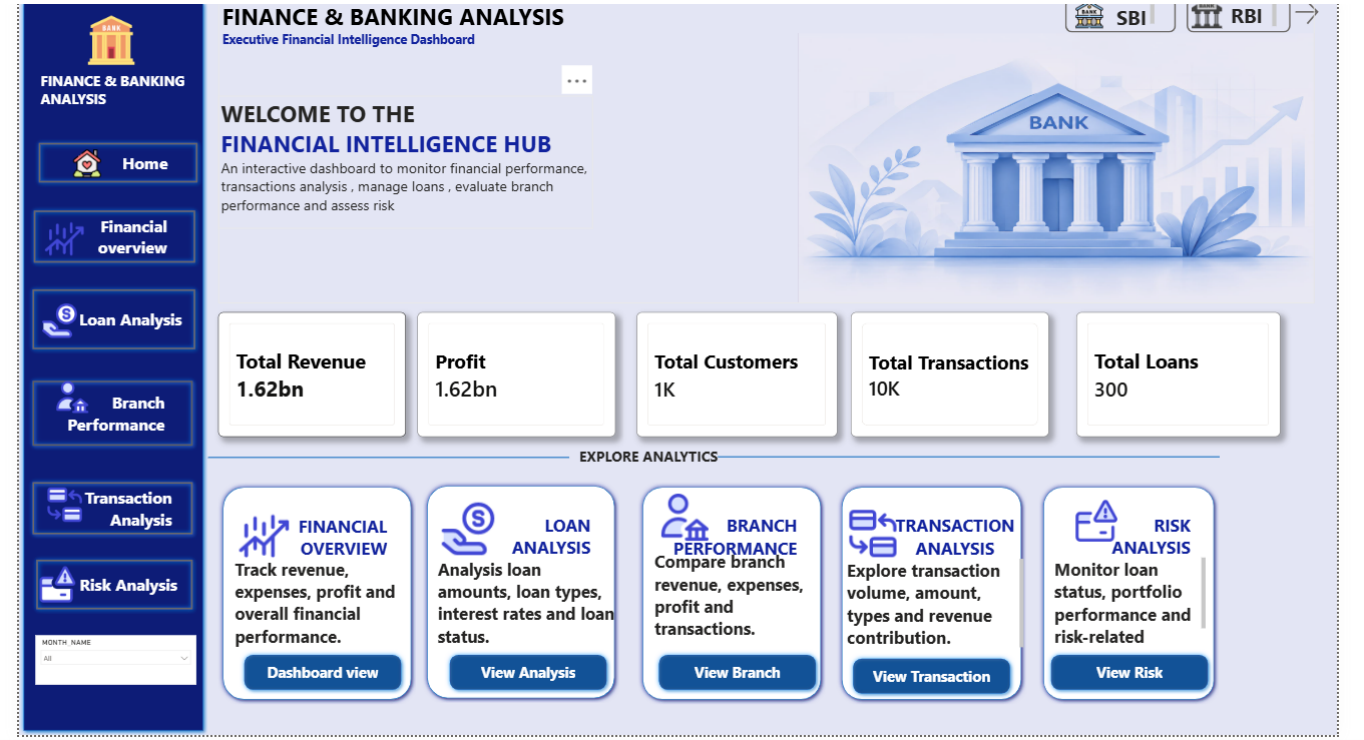
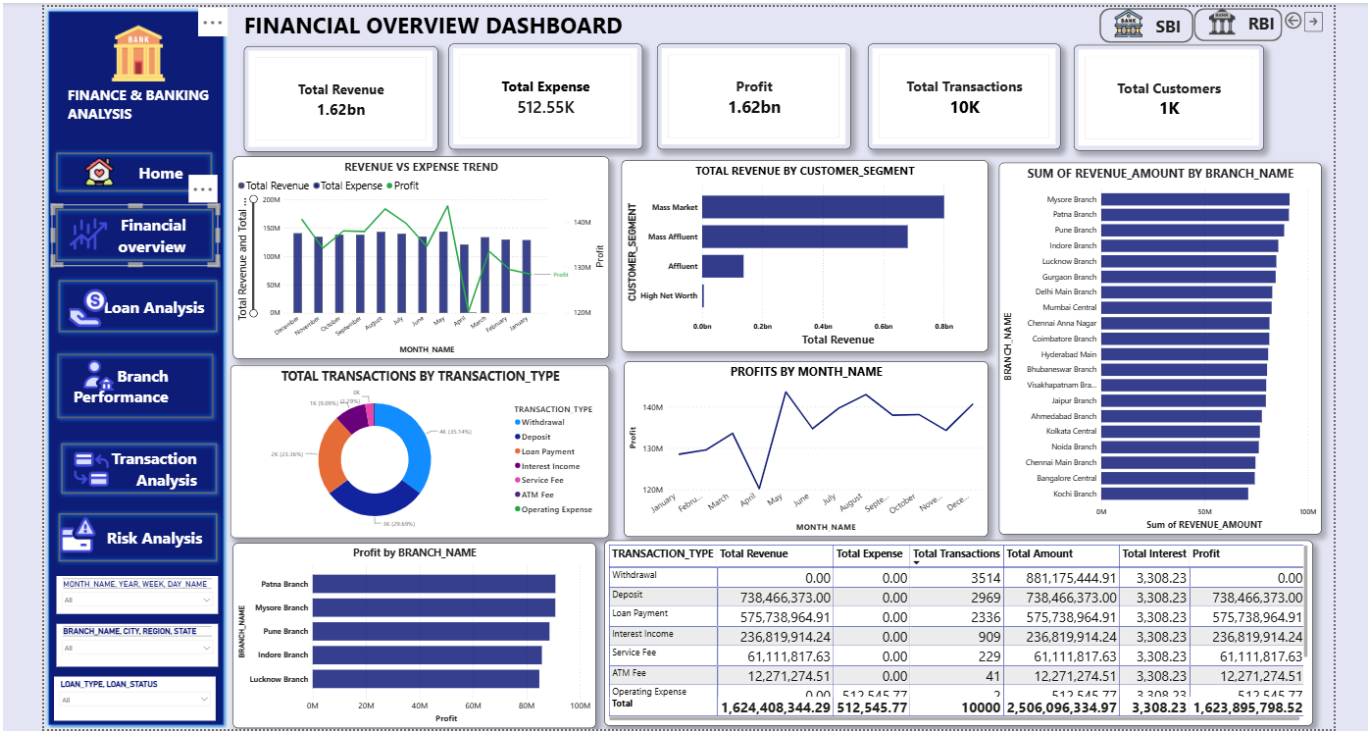


Figure 1. Power BI Home Page – navigation cards for dashboard sections.

The Home Page provides an overview of the Finance Performance and Banking Analysis Dashboard. It contains navigation buttons and key highlights that allow users to easily access different analytical pages and explore financial, loan, branch, transaction, and risk-related information.

8. User Guide Screenshot

8.2 Financial Overview



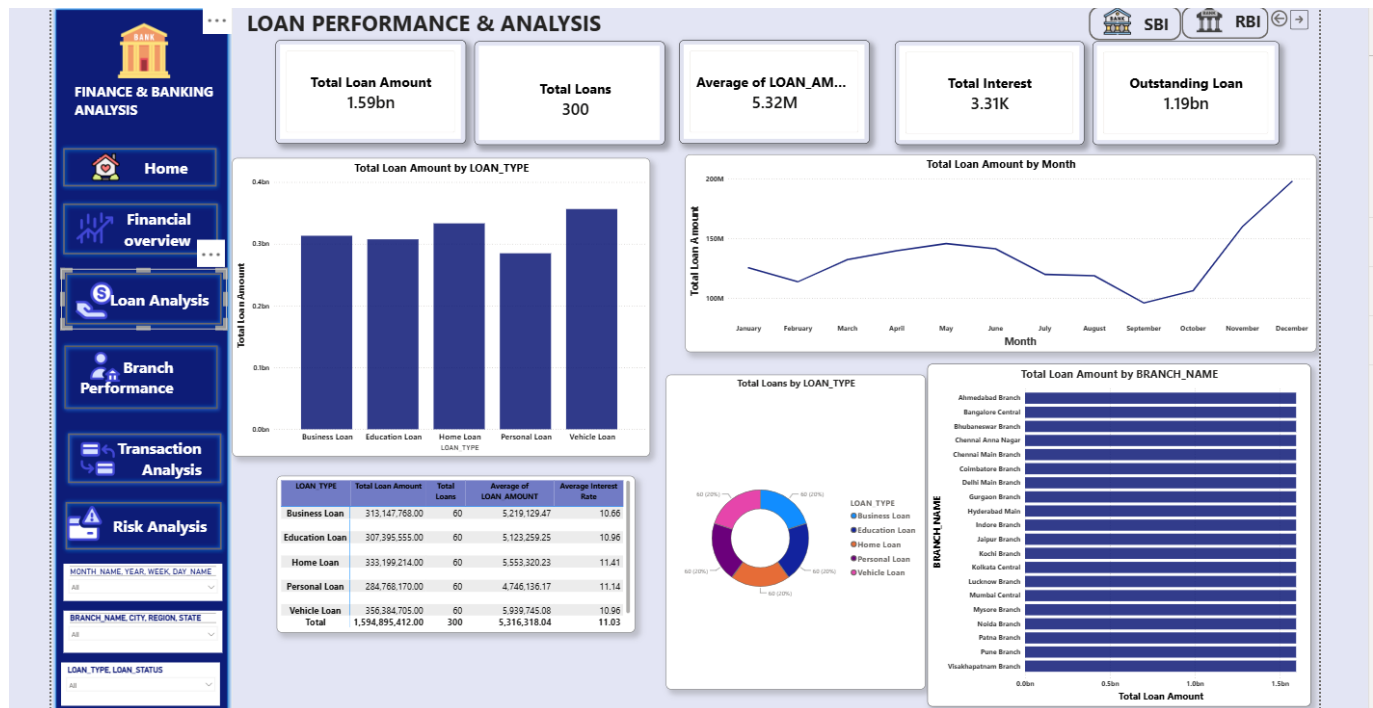
Review revenue, expenses, profit, profit margin and transaction KPIs. Use branch, date and transaction filters to explore the data.

The **Financial Overview page** is designed as a single interactive dashboard using **Power BI bookmarks**. Instead of creating separate pages for each analysis, multiple bookmarks are used to display different analytical views within the same page. Users can select the required bookmark to switch between different areas of banking analysis.

Bookmarks included:

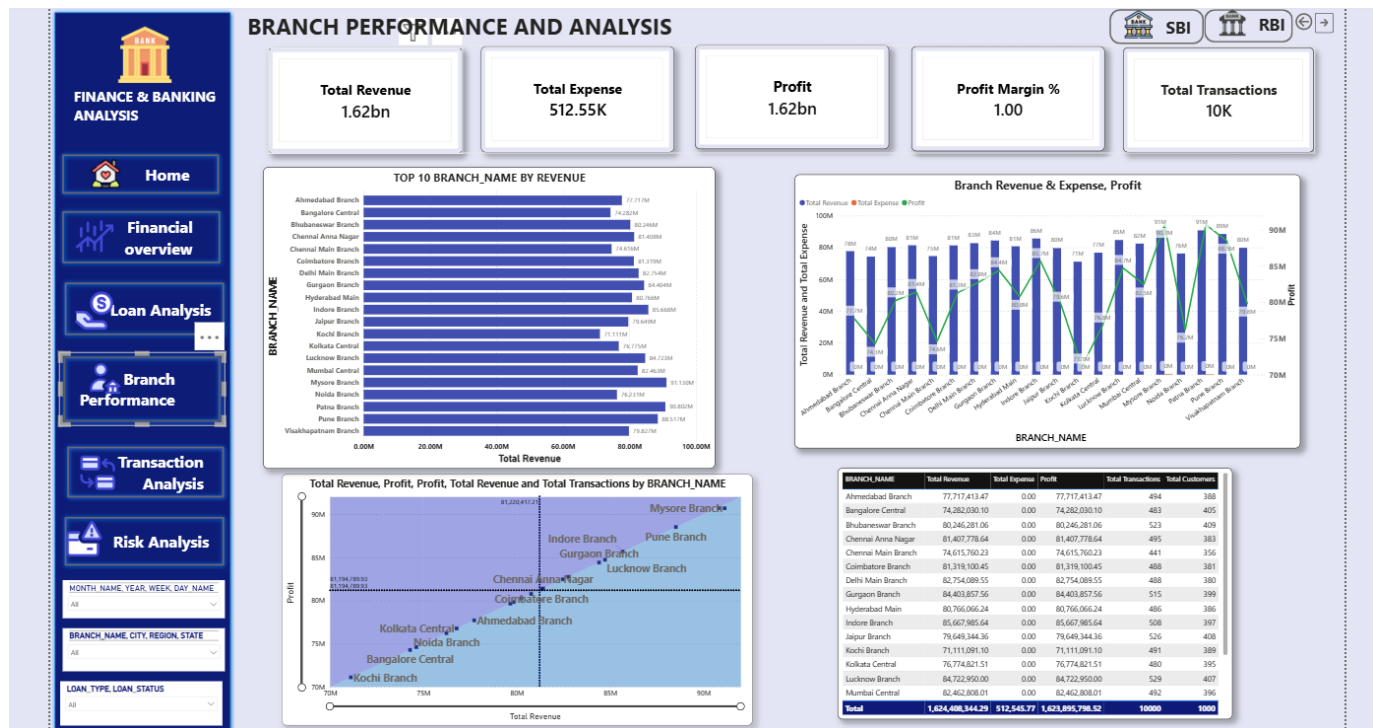
- **Financial Overview** – Shows overall financial performance and key KPIs.
- **Loan Analysis** – Shows loan portfolio, loan types, loan status, and interest-rate analysis.
- **Branch Analysis** – Shows branch-wise revenue, expenses, profit, and performance.
- **Transaction Analysis** – Shows transaction volume, transaction amount, transaction types, and revenue.
- **Risk Analysis** – Shows loan status and risk-related portfolio information

8.3 Loan Analysis



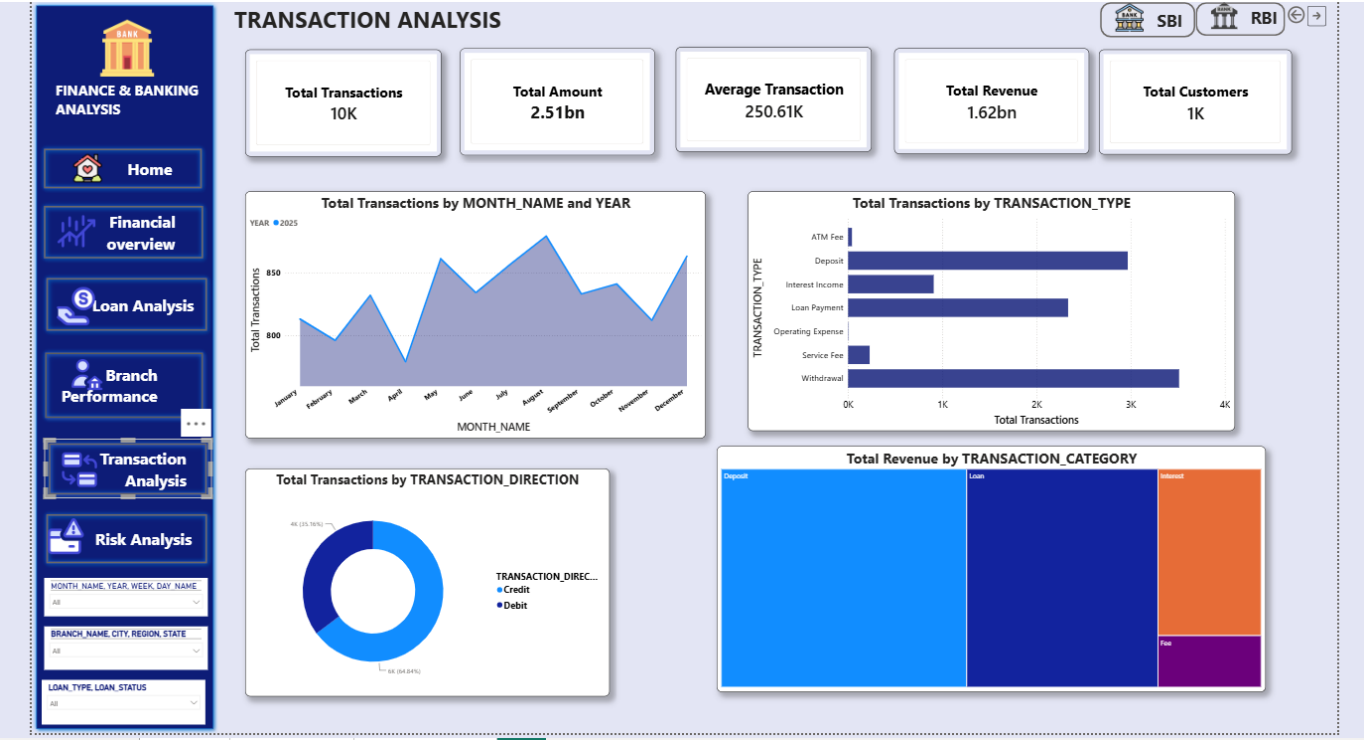
Use loan type, status, branch and date-related filters to analyze the loan portfolio.

8.4 Branch Performance



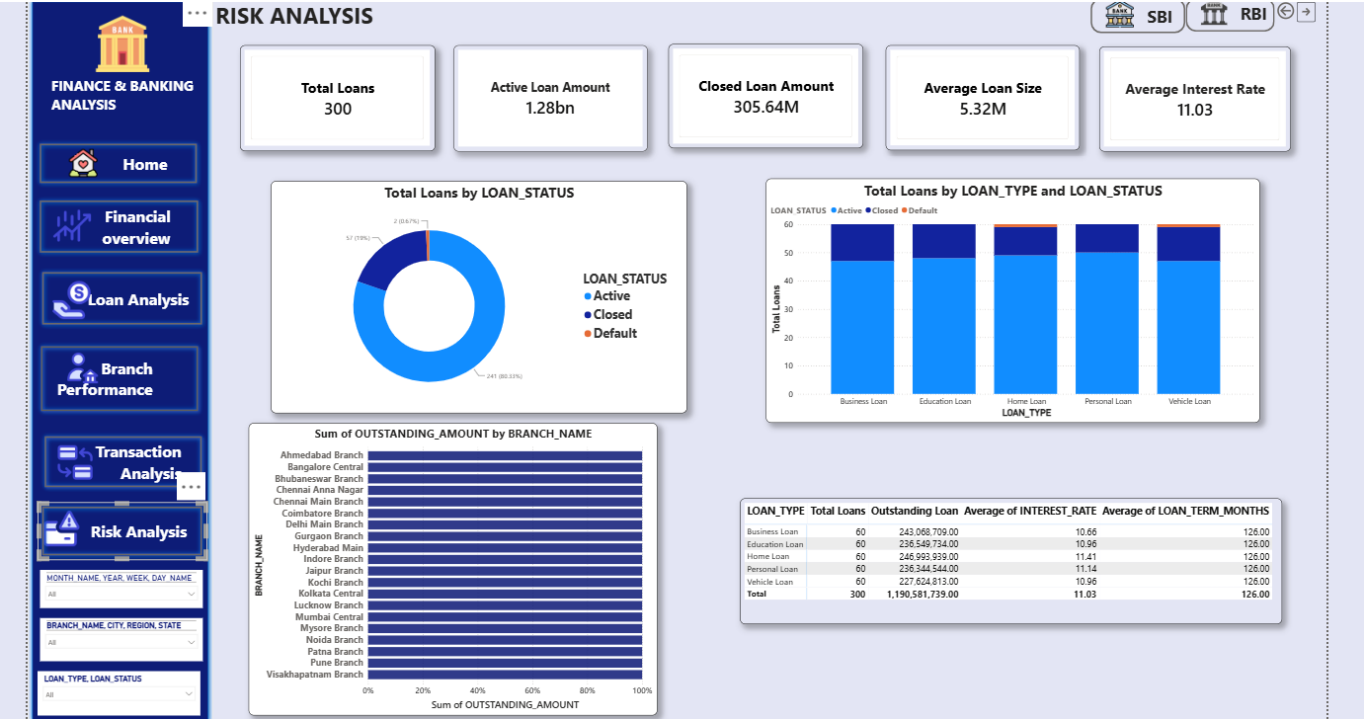
Compare branch-level revenue, expense, profit and transaction activity. Use the branch visual to drill through to Branch Details.

8.5 Transaction Analysis



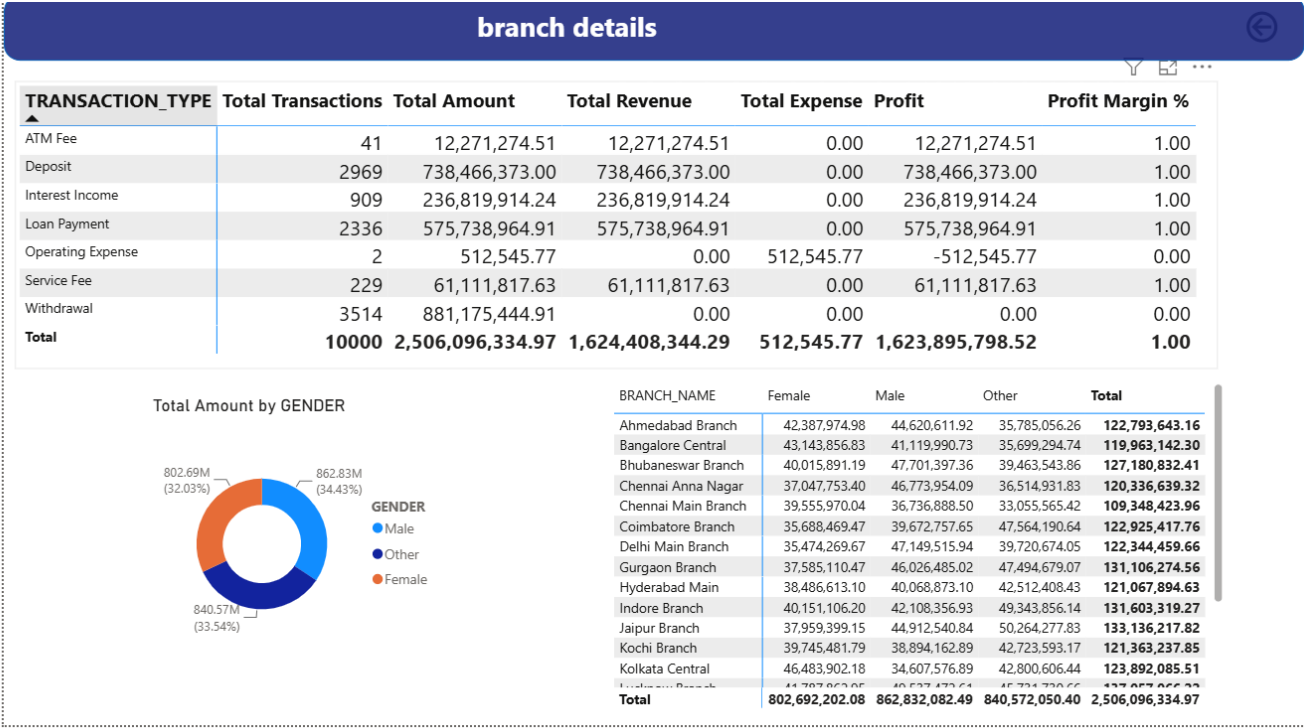
Analyze transaction volume and value by transaction type, category and direction.

8.6 Risk Analysis



Review loan status distribution, loan portfolio and interest-rate-related indicators.

8.7 Branch Details Drillthrough



Right-click a branch in a supported branch visual and select Drillthrough → Branch Details. The selected branch is passed automatically to the detail page.

The Branch Details page contains a Matrix with transaction type and financial measures.

9. Dashboard Features

Feature	Implementation
Page Navigation	Buttons on the Home page navigate to report pages.
Slicers	Branch, loan, date and transaction attributes can be used for filtering.
Drillthrough	Branch Name is used to pass the selected branch to Branch Details.
Bookmarks	Can be used to create alternate views or show/hide report elements.
Tooltips	Additional KPIs can be displayed when users hover over visuals.
Web URL	A button can open an external official website when required.
RLS	User access can be restricted using a dynamic branch-based security rule.

10. Column Data Dictionary

10.1 Fact Table – FACT_FINANCIAL_TRANSACTION

Column	Description
TRANSACTION_KEY	Unique key for the transaction record.
TRANSACTION_ID	Business transaction identifier.
DATE_KEY	Links the transaction to the date dimension.
BRANCH_KEY	Links the transaction to the branch dimension.
CUSTOMER_KEY	Links the transaction to the customer dimension.
ACCOUNT_KEY	Links the transaction to the account dimension.
LOAN_KEY	Links the transaction to the loan dimension where applicable.
TRANSACTION_TYPE_KEY	Links the transaction to DIM_TRANSACTION.
TRANSACTION_AMOUNT	Amount associated with the transaction.
REVENUE_AMOUNT	Revenue generated by the transaction.
EXPENSE_AMOUNT	Expense associated with the transaction.
BALANCE_AFTER_TRANSACTION	Account balance after the transaction.

10.2 Dimension Tables

Table	Important Columns
DIM_DATE	FULL_DATE, DAY, MONTH, MONTH_NAME, QUARTER, YEAR, WEEK, DAY_NAME, IS_WEEKEND
DIM_BRANCH	BRANCH_KEY, BRANCH_NAME and branch attributes
DIM_TRANSACTION	TRANSACTION_TYPE_KEY, TRANSACTION_TYPE, TRANSACTION_CATEGORY, TRANSACTION_DIRECTION, REVENUE_FLAG, EXPENSE_FLAG
DIM_LOAN	LOAN_KEY, LOAN_ID, CUSTOMER_KEY, LOAN_AMOUNT, LOAN_TYPE, LOAN_STATUS, INTEREST_RATE, LOAN_START_DATE, LOAN_TERM_MONTHS
DIM_CUSTOMER	Customer attributes including GENDER where available
DIM_ACCOUNT	Account-level attributes

11. DAX Measures

11.1 Financial Measures

Total Revenue

Total Revenue =
SUM(FACT_FINANCIAL_TRANSACTION[REVENUE_AMOUNT])

Total Expense

Total Expense =
SUM(FACT_FINANCIAL_TRANSACTION[EXPENSE_AMOUNT])

Profit

Profit =
[Total Revenue] - [Total Expense]

Profit Margin %

```
Profit Margin % =  
DIVIDE([Profit], [Total Revenue], 0)
```

11.2 Transaction Measures

Total Transactions

```
Total Transactions =  
COUNTROWS(FACT_FINANCIAL_TRANSACTION)
```

Total Amount

```
Total Amount =  
SUM(FACT_FINANCIAL_TRANSACTION[TRANSACTION_AMOUNT])
```

Average Transaction Amount

```
Average Transaction Amount =  
DIVIDE([Total Amount], [Total Transactions], 0)
```

Revenue per Transaction

```
Revenue per Transaction =  
DIVIDE([Total Revenue], [Total Transactions], 0)
```

11.3 Loan Measures

Total Loans

```
Total Loans =  
DISTINCTCOUNT(DIM_LOAN[LOAN_ID])
```

Total Loan Amount

```
Total Loan Amount =  
SUM(DIM_LOAN[LOAN_AMOUNT])
```

Average Loan Amount

```
Average Loan Amount =  
AVERAGE(DIM_LOAN[LOAN_AMOUNT])
```

Average Interest Rate

```
Average Interest Rate =  
AVERAGE(DIM_LOAN[INTEREST_RATE])
```

Active Loans

```
Active Loans =  
CALCULATE([Total Loans], DIM_LOAN[LOAN_STATUS] = "Active")
```

Closed Loans

```
Closed Loans =  
CALCULATE([Total Loans], DIM_LOAN[LOAN_STATUS] = "Closed")
```

11.4 Time Intelligence

Previous Month Revenue

```
Previous Month Revenue =  
CALCULATE([Total Revenue], DATEADD(DIM_DATE[FULL_DATE], -1, MONTH))
```

MoM Revenue Change

```
MoM Revenue Change =  
[Total Revenue] - [Previous Month Revenue]
```

MoM Revenue %

```
MoM Revenue % =  
DIVIDE([MoM Revenue Change], [Previous Month Revenue], 0)
```

12. Drill through & Branch Details

The Branch Details page is designed as a focused drillthrough page. The Branch Name field from DIM_BRANCH is placed in the Drill through field well.

12.1 Branch Details Matrix

Matrix Area	Field / Measure
Rows	DIM_TRANSACTION[TRANSACTION_TYPE]
Values	[Total Transactions]
Values	[Total Amount]
Values	[Total Revenue]
Values	[Total Expense]
Values	[Profit]
Values	[Profit Margin %]

When the user drills through from a branch visual, the selected branch is automatically applied as the drill through filter.

13. Row-Level Security (RLS)

Dynamic RLS can be implemented using an RLS_UserAccess table containing userEmail, Role and Branch. An ALL value can be used for users who should access all branches.

UserEmail	Role	Branch
admin@example.com	Admin	ALL
viewer1@example.com	Viewer	BR001
viewer2@example.com	Viewer	BR002
viewer3@example.com	Viewer	BR003

Dynamic_RLS filter on DIM_BRANCH:

```
VAR CurrentUser =  
    USERPRINCIPALNAME()
```

```

VAR UserBranch =
    LOOKUPVALUE(
        RLS_UserAccess[Branch],
        RLS_UserAccess[UserEmail],
        CurrentUser
    )

RETURN
    UserBranch = "ALL"
    || DIM_BRANCH[BRANCH_ID] = UserBranch

```

After creating the role, test it in Power BI Desktop using Modeling → View as. After publishing, assign users to the role in Power BI Service.

14. Testing & Validation

Test	Expected Result
Home navigation	Each navigation button opens the correct report page.
Branch filter	Selecting a branch filters related visuals.
Drill-through	Selected branch is passed automatically to Branch Details.
Matrix totals	Matrix totals agree with the corresponding measures.
Loan filters	Loan type/status filters update loan visuals.
Deposit gender analysis	Gender donut shows deposit amount when transaction type is filtered to Deposit.
RLS	Users only see data allowed by their assigned branch/access level.

15. Conclusion

The Finance & Banking Analysis Power BI project provides an interactive reporting solution for understanding financial performance, transactions, branches, customers and loans. The combination of Snowflake, Power BI and DAX enables users to move from high-level KPIs to detailed branch-level analysis while maintaining an organized and secure reporting structure.

16. Future Enhancements

- Add automated data refresh through Power BI Service.
- Add additional customer segmentation and behavioral analysis.

- Introduce more advanced loan risk indicators when validated risk fields are available.
- Add forecasting for revenue and loan portfolio trends.
- Add mobile-optimized report layouts.
- Extend RLS to support multiple branches per user where required.